

The Cost of National Pricing: Side Payments and Efficiency under Self-Dispatch with Redispatch in the GB Electricity Market

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Abstract

Liberalised electricity markets resolve the nonconvexities created by generation commitment through out-of-market side payments whose magnitude and opacity concern regulators. Most analyses adopt the centralised, nodal-dispatch institution of US independent system operators (ISOs), in which these appear as make-whole payments on top of locational prices. Great Britain (GB) occupies the opposite institutional corner: a single national uniform price cleared under self-dispatch, with network constraints and locked-in commitment resolved afterwards through redispatch in the Balancing Mechanism, the net cost of which is *socialised* – recovered uniformly from demand through balancing charges rather than charged to the parties that cause it. Following the 2025 Review of Electricity Market Arrangements (REMA), which retained national pricing and rejected locational pricing, the policy-relevant question is the cost of that choice. We formalise GB arrangements as a two-stage market – a network-blind national clearing followed by a cost-minimising redispatch – and prove two central results. First, in a convex economy national-pricing-with-redispatch is welfare-equivalent to nodal pricing, so every unit of realised welfare loss is attributable to commitment nonconvexity. Second, the visible cost of national pricing decomposes into a congestion-rent component that nodal pricing would internalise as merchandising surplus (the standard term for the network revenue an ISO collects from pricing energy differently on the two sides of a binding constraint), plus a strategic redispatch markup; we show this gap is non-negative under the empirically supported condition that minimised nodal make-whole payments are small. We quantify the gap on the Scotland–England (B6) boundary using public Balancing Mechanism data, exploiting the exact correspondence between the model’s two stages and the settlement objects (final physical notifications and flagged bid-offer acceptances), and validate the figure against the published constraint cost. The cost of national pricing is, to first order, the same congestion rent: under nodal pricing the system operator would *collect* it as merchandising surplus (and could rebate it to consumers), whereas under national pricing it is instead *paid out* to generators as redispatch and socialised across demand. Measured directly from settlement records, this realised B6 congestion rent is of order £1.1bn in FY 2024/25 – about 59% of GB system constraint costs and tracking the independent published daily series almost exactly (correlation 0.97); the underlying 9.55 TWh of Scottish-wind curtailment matches independently reported volumes. Table 3 summarises the per-period magnitude on the B6 boundary.

Keywords: Electricity market design, National pricing, Redispatch, Congestion rent, Balancing mechanism, Nonconvex pricing

1. Introduction

Wholesale electricity markets must allocate, over a constrained transmission network, a good that must be produced and consumed in instantaneous balance. Electrical *energy* can of course be stored – in batteries, pumped hydro or as a fuel – but only at limited scale and non-trivial cost, so in operational timescales generation must still track demand period by period. Participants on the supply side face nonconvex costs: start-up and no-load costs, minimum-load and minimum-up/down-time constraints, and ramping limits. Example 1 in §3.1 depicts a two-zone numerical illustration, carried through the whole paper, of how these nonconvexities and a binding network constraint together generate the side payments we study. A long line of results establishes that, under such nonconvexities, a Walrasian equilibrium supported by linear and anonymous prices need not exist [30, 6, 7]. Market operators therefore relax one or more design desiderata – efficiency, individual rationality, budget balance, or envy-freeness – and compensate the resulting losses through out-of-market side payments. Because these payments are not embedded in the public price, they blunt the market’s scarcity signal and distort entry, exit and investment incentives, which is why they have attracted sustained regulatory scrutiny [22, 5].

The operational-research and power-systems literatures have largely studied this problem within one institution: the centralised, nodal-dispatch market of the US independent system operators (ISOs). There, the operator solves a security-constrained unit commitment problem to fix the efficient dispatch and then derives linear, anonymous *locational marginal prices*; because these prices need not be individually rational, generators receive discriminatory *make-whole* (uplift) payments. A substantial body of work designs prices that minimise these payments: integer-programming pricing [30], extended locational marginal pricing [32], and convex hull pricing [14], which minimises total uplift by pricing on the convex envelope of the commitment problem [17, 20, 3]. Bichler et al. [5] bring this question into the market design perspective, treating efficiency and individual rationality as first-order goals and budget balance and envy-freeness as second-order, and proposing pricing rules (PBE-A, PE-A) that minimise make-whole payments while retaining linear, anonymous prices. In our paper we build upon these pricing rules, adapting them to the Great Britain context.

Great Britain (GB) occupies the opposite institutional corner, to a nodal-dispatch model. GB clears a *single national* uniform price per half-hourly settlement period under *self-dispatch*: generators and suppliers contract bilaterally and through day-ahead and intraday auctions, and declare a physical position – the Final Physical Notification (FPN) at gate closure. This is a half-hourly MW profile each balancing mechanism unit expects to deliver, as defined in the Balancing and Settlement Code and published by Elexon [11, 12]. The transmission network is invisible to

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this clearing. The National Energy System Operator (NESO) then makes the schedule physically feasible *ex post* through *redispatch* in the Balancing Mechanism (BM), accepting personalised, pay-as-bid offers (to turn up) and bids (to turn down) and socialising the net cost across demand through balancing charges. The scale is already material: on high-wind days NESO routinely pays Scottish wind farms to switch off (curtailment) behind the B6 boundary while paying English gas plant a premium to make up the shortfall – curtailment payments alone exceeded £1bn in 2024 and single-day constraint costs have topped £10m [28, 25]. The side-payment problem does not disappear under national pricing; it is relocated from make-whole payments on locational prices into socialised redispatch cost on a single price – and rendered *less* transparent, because the cost is not even attributed to identified units.

This institutional difference has become acutely policy-relevant. The 2025 Review of Electricity Market Arrangements (REMA) concluded a three-year debate by rejecting both zonal and nodal pricing and committing GB to a reformed national-pricing model [9]. The decision was contested precisely because the quantitative case turned on disputed estimates of the dispatch efficiency that locational pricing would unlock, with industry studies reaching materially different conclusions about the magnitude of that efficiency gain – the “size of the prize”, in the language of the policy debate [1, 34]. Meanwhile the cost being argued over is large and rising: transmission constraint costs reached roughly £1.9bn in 2024/25 – about 71% of GB balancing costs, up from 44% in 2022/23 – and are projected at £4–8bn per year by 2030 [28, 26], overwhelmingly driven by curtailment of Scottish wind behind the Scotland–England boundary, with replacement gas historically carrying a premium of around 30% over wholesale prices [28, 25]. The relevant question is therefore no longer whether GB should price locationally, but what retaining national pricing costs, and what reduces it.

A recurring concern about any uniform-price-plus-redispatch design is that it invites strategic conduct. Under *increase–decrease* (inc-dec) gaming, generators anticipate the redispatch stage and bid strategically in the national market – bidding low at export-constrained locations to be scheduled and then paid to turn down, or high to be paid to turn up – so that the redispatch markup reflects market design rather than market power [16, 19]. This literature has developed largely as stylised game-theoretic analysis on small networks [18]; it motivates our treatment of the redispatch markup as an object to be *estimated* from data rather than assumed.

Contribution. We make five contributions. First, we formalise GB arrangements as a two-stage market – a network-blind national clearing followed by a cost-minimising redispatch – and show that national pricing is the maximal envy-freeness-relaxation point of the design framework of Bichler et al. [5]: it abandons locational envy-freeness entirely and restores physical feasibility through redispatch. Second, we prove (Proposition 1) that in a convex economy national-pricing-with-redispatch is welfare-equivalent to nodal pricing, so that every unit of realised welfare loss is attributable to commitment nonconvexity; this turns “national pricing is inefficient” into a testable claim. Third, considering a congestion-rent term as the value of the scarce transfer capacity; we prove (Proposition 2) that the visible cost of national pricing decomposes into this congestion-rent term and a strategic redispatch markup. We also show that the gap relative to minimised nodal make-whole payments is non-negative under a condition supported by

the near-zero PE-A payments reported by Bichler et al. [5]. Fourth, we quantify this gap on the Scotland–England (B6) boundary using public Balancing Mechanism data, exploiting an exact correspondence between the model’s two stages and the GB settlement objects, and we validate the headline figure by reconciling it with NESO’s published constraint cost; measured directly from settlement records the realised B6 congestion rent is of order £1.1bn in FY 2024/25, about 59% of GB system constraint cost. Fifth, we cast the placement of flexibility across the nested GB boundaries as a siting-and-operation optimisation (Proposition 3) and show its marginal value rises up the boundary cascade, so the budget should water-fill the most upstream binding location first – turning “strategic siting” into a ranked, quantified prescription. The approach is not specific to electricity: it applies to any two-stage market that clears at a uniform price and resolves feasibility through redispatch. All models, data loaders and experiment scripts are released as an open repository accompanying the submission, so every figure and table is reproducible from public data (see the reproducibility note).

2. Related literature and positioning

Beyond the GB-specific REMA literature discussed below, a parallel European stream studies the same uniform-pricing-plus-redispatch architecture under the EU zonal model: the efficiency and gaming consequences of zonal versus nodal configurations [10, 2], the design of flow-based market coupling [4], and the welfare effects of bidding-zone reconfiguration [33]. Recent GB-focused work – on the system value of locational signals and the integration of high renewable shares under a national price [29, 15] – motivates the cost-of-the-status-quo question we make precise. We position against both streams throughout.

Pricing in nonconvex markets. The foundational difficulty is that, with indivisibilities and nonconvex costs, competitive-equilibrium prices may be nonlinear and personalised, or may fail to exist [7, 6]. Because linear, anonymous prices are nonetheless essential as a baseline for forward and derivative markets [8], practical designs retain them and add discriminatory uplift. The dominant approaches – integer-programming pricing [30], extended locational marginal pricing [32], and convex hull (minimum-uplift) pricing [17, 14] – all amount to pricing on a convex relaxation of the commitment problem and differ in how they compute the dual prices and uplift. Computing convex hull prices efficiently is itself an active area [20, 3], and the equilibrium interpretation of the required compensation has been formalised as a binary equilibrium problem with compensation [21]. Liberopoulos and Andrianesis [22] provide the standard critical review, and Bichler et al. [5], whose make-whole-minimising PE-A/PBE-A rules we extend, recast the problem as one of market design with budget balance prioritised over envy-freeness. All of this work assumes a single, centrally dispatched market which differ from the GB case; nonetheless we retain its objects (DCOPF, make-whole minimisation, convex-hull relaxation for nonconvex shadow prices) but embed them in a two-stage, self-dispatch-then-redispatch mechanism.

Locational versus uniform pricing and redispatch. A parallel economics literature compares nodal, zonal and uniform pricing with ex-post redispatch. Its central cautionary result is inc-dec gaming: when a uniform-price market is

followed by a locational redispatch market, generators bid strategically in the first stage to profit in the second, so that redispatch cost can reflect inconsistent market design rather than fundamentals or even classical market power [19, 16, 18]. These analyses are typically game-theoretic on small, stylised networks [19, 16]. We complement them in two ways: our Proposition 1 isolates, on a full DCOPF, the portion of the welfare gap that is intrinsic (commitment nonconvexity) from the portion that is mechanism-induced, and our empirical design *estimates* the redispatch markup – the footprint of such conduct – rather than positing it, finding (via a within-unit-day regression discontinuity, §5) no significant boundary markup at B6, so that the realised cost is congestion rent rather than gaming.

GB market design and the REMA debate. The reform debate has produced a wave of quantitative GB studies, including detailed locational-pricing assessments and welfare estimates whose magnitudes are disputed [1, 34, 13]. Several use a reduced GB transmission representation driven by the Future Energy Scenarios [27] to compute zonal or nodal prices, an approach we adopt for external validity. This literature is predominantly consultancy and policy analysis – for instance the modelling commissioned during REMA by AFRY [1], and the UKERC and Energy Systems Catapult assessments [34, 24, 13] – oriented to the *ex ante* efficiency case for changing the wholesale market; comparatively little of it is framed in terms of the side-payment and pricing-rule theory above, and almost none exploits the settlement-level Balancing Mechanism record to measure the realised cost of the status quo. Our contribution is to supply that bridge: a theory of the cost of national pricing, and a data-driven measurement of it.

The GB electricity market neither dispatches centrally nor prices locationally, so the side-payment problem migrates from make-whole payments on nodal prices into socialised redispatch cost on a single national price. We show this is not outside the Bichler et al. [5] framework but an extreme point of it: their framework grades market designs by how many of the desiderata (linear-anonymous prices, individual rationality, budget balance, envy-freeness) are retained, and national pricing is the design that gives up *locational* envy-freeness in full – a single price treats a constrained Scottish and an unconstrained English generator alike – restoring feasibility through redispatch instead. Within that framework, Bichler et al. [5] show that demand-side flexibility lowers the make-whole payments a nodal market must pay; we carry the same mechanism up to the system level, where reducing the surplus behind a binding boundary is precisely how REMA’s reformed national pricing hopes to contain constraint costs (made quantitative in §6.4).

3. A two-stage model of national pricing with redispatch

This section develops the model in full. We first fix the model primitives – the exogenous objects from which everything else is built: the zones and settlement periods, the generators with their costs and technical constraints, the buyers with their valuations, and the network – together with the boundary-transfer representation of that network (§3.1), then state the three optimisation programs that the analysis compares: the nodal benchmark (§3.2), the network-blind national clearing that fixes the self-schedule and the single price (§3.3), and the redispatch that restores

feasibility (§3.4). We then show that the two stages correspond exactly to observable GB settlement objects (§3.5), and locate national pricing within the design framework of Bichler et al. [5] (§3.6).

3.1. Primitives and the boundary-transfer network

We consider a system that has zones $n \in \mathcal{N}$ and settlement periods $t \in \mathcal{T}$; the traded goods are the pairs $\mathcal{N} \times \mathcal{T}$. Generators $j \in \mathcal{J}$ sit at a zone, with $\mathcal{J}_n$ the generators in zone n ; buyers $i \in \mathcal{I}$ likewise, with $\mathcal{I}_n$ the buyers in zone n . Each generator has a per-period output $y_{jt} \geq 0$, a commitment indicator $u_{jt} \in \{0, 1\}$, and a start-up indicator $d_{jt} \in \{0, 1\}$, with variable cost c_{jt} and no-load/fixed cost h_{jt} incurred when committed. Technical constraints comprise minimum and maximum stable load, $\underline{P}_j u_{jt} \leq y_{jt} \leq \bar{P}_j u_{jt}$, ramp limits $|y_{jt} - y_{j,t-1}| \leq R_j$, and minimum up/down times encoded through (u, d) ; we write these collectively as $(y, u, d) \in \mathcal{T}_j$. Buyer i consumes $x_{it} \in [0, \bar{D}_{it}]$ with marginal valuation v_{it} ; a buyer is *price-inelastic* (must-serve) when its demand must be met at any price, and *price-sensitive* otherwise. Welfare is $W = \sum_{i,t} v_{it} x_{it} - \sum_{j,t} (c_{jt} y_{jt} + h_{jt} u_{jt})$.

The *net injection* of zone n in period t is

$$p_{nt} = \sum_{j \in \mathcal{J}_n} y_{jt} - \sum_{i \in \mathcal{I}_n} x_{it}. \quad (1)$$

Zones are connected by inter-zonal corridors $\ell \in \mathcal{L}$, each a boundary (cut) in the transmission network; $f_{\ell t}$ is the corridor flow and $A_{n\ell} \in \{-1, 0, +1\}$ the zone–corridor incidence (with $A_{n\ell} = +1/-1$ when ℓ exports/imports from/to n). The corridor capability $\bar{B}_{\ell t}$ is the boundary limit published in NESO’s Electricity Ten Year Statement (ETYS), the annual statement of GB transmission boundary transfer capabilities. We use this *boundary-transfer* (net-transfer-capacity) representation, rather than per-line flows, because GB constraints are defined and managed at boundary granularity – the Scotland–England corridor B6 being the binding example – and ETYS publishes capabilities at exactly this level. Equivalently, a linearised DC representation would replace the transport constraints below with $f_{\ell t} = \sum_n \Phi_{\ell n} p_{nt}$ for a power-transfer-distribution-factor matrix Φ ; the boundary form is the natural reduction for the GB study and keeps the programs small enough to solve period-by-period at half-hourly resolution.

Figure 1 sketches the resulting picture for GB: cheap wind in the Scottish zones is a net exporter, demand-heavy English zones are net importers, and the B6 corridor between them binds.

Example 1 (The B6 boundary, carried through the paper). A single settlement period, two zones sco (Scotland) and ENG (England) joined by the B6 corridor with capability $\bar{B} = 500$ MW. Demand is inelastic: 100 MW in sco, 900 MW in ENG. Two generators: Scottish wind with cost $c_w = £2/\text{MWh}$ and capacity 800 MW, and English gas with cost $c_g = £60/\text{MWh}$ and ample capacity. *Stage 1* (network-blind, Problem (3)) clears the cheapest mix that meets the 1000 MW total: wind 800, gas 200, setting the national price $\lambda^{\text{nat}} = £60/\text{MWh}$. But this self-schedule sends $800 - 100 = 700$ MW across B6, exceeding the 500 MW capability by $s = 200$ MW, so it is *infeasible* for the network of Problem (2). *Stage 2* (redispatch, Problem (4)) restores feasibility at least cost by turning wind *down* 200 MW and gas *up* 200 MW. At cost-reflective prices the redispatch cost is $RC = c_g \cdot 200 - c_w \cdot 200 = £11,600$, which equals the

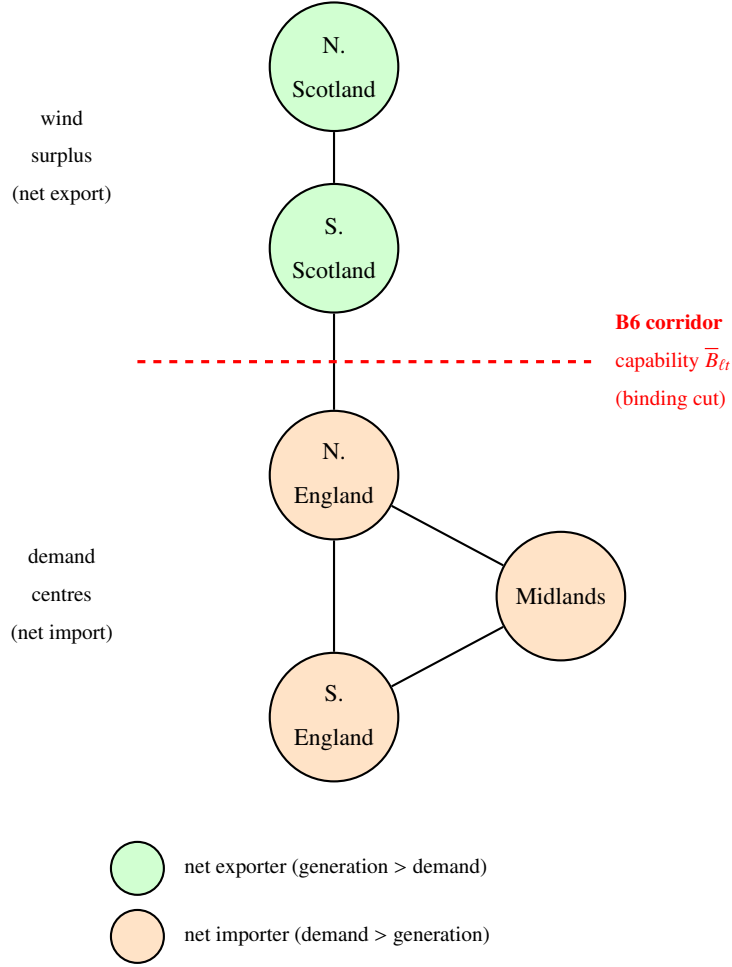


Figure 1: Stylised GB zonal network. Scottish zones export cheap wind across the B6 corridor to demand-heavy English zones; when the ETYS capability $\bar{B}_{\ell t}$ binds, the network-blind self-schedule is infeasible and must be redispatched. Node area is indicative only; the geographically resolved full and B6-reduced networks are shown in Figure 2.

congestion rent $R_{\text{cong}} = \pi s$ with boundary shadow price $\pi = c_g - c_w = \text{£}58/\text{MWh}$ and relieved overload $s = 200 \text{ MW}$ (Proposition 2). Because the economy here is convex, the realised dispatch coincides with the nodal optimum, no make-whole payment is needed, and the entire cost of national pricing is this $\text{£}11,600$ of congestion rent – collected as merchandising surplus under nodal pricing, but paid out to generators and socialised under national pricing. Adding a strategic turn-up markup μ on the gas offer raises RC by $M = \mu c_g \cdot 200$ (e.g. $+\text{£}3,600$ at $\mu = 0.30$); replacing the convex wind unit with a committable gas unit makes Stage 1 commit blind to the network and opens the welfare gap of Proposition 1(b). These are exactly the `synthetic_b6` and `synthetic_b6_nonconvex` instances solved by `gb_two_stage_skeleton.py` in the repository.

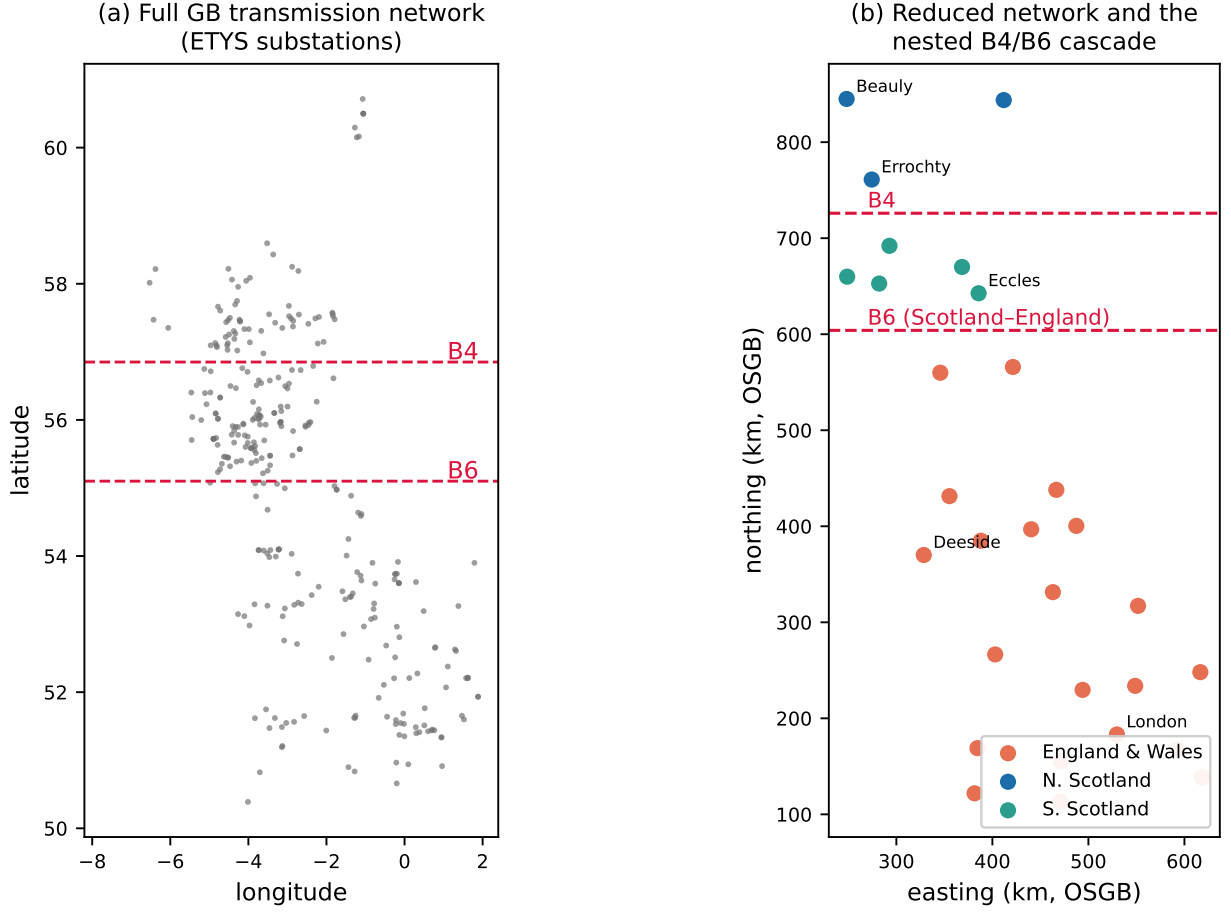


Figure 2: The GB transmission network and its reduction, from the open PyPSA-GB data [23]. (a) the full network: ETYS substation locations, which trace GB, with the nested B4 and B6 boundaries marked. (b) the reduced 29-bus network (OSGB coordinates), coloured by the three model zones – North Scotland, South Scotland and England & Wales – separated by the nested cuts B4 and B6 used by the cascade model of §4.1; the headline B6 study aggregates the two Scottish zones into a single sco north of B6. Generated by `make_gbmap.py`.

3.2. The nodal benchmark

The network-feasible efficient dispatch maximises welfare subject to zonal balance, corridor limits and the generators’ technical sets:

$$\begin{aligned}
 \max_{x,y,u,d,f} \quad & \sum_{i,t} v_{it} x_{it} - \sum_{j,t} (c_{jt} y_{jt} + h_{jt} u_{jt}) \\
 \text{s.t.} \quad & p_{nt} = \sum_{\ell} A_{n\ell} f_{\ell t} & \forall n, t \quad (\lambda_{nt}) \\
 & -\bar{B}_{\ell t} \leq f_{\ell t} \leq \bar{B}_{\ell t} & \forall \ell, t \quad (\pi_{\ell t}) \\
 & (y, u, d) \in \mathcal{T}_j, \quad 0 \leq x_{it} \leq \bar{D}_{it}, \quad u_{jt} \in \{0, 1\}.
 \end{aligned} \tag{2}$$

Its optimum is (x^*, y^*, u^*) with welfare W^* . With the binary commitment variables $u_{jt} \in \{0, 1\}$ present, (2) is a mixed-integer program and the parenthetical multipliers $\lambda_{nt}, \pi_{\ell t}$ are *not* the duals of Problem (2) itself – a nonconvex program has no exact dual prices. As is standard in nonconvex electricity pricing [30, 14, 22], the prices are defined on a convex restriction or relaxation: fix the integers at the optimal commitment $u = u^*$ and take the duals of the resulting linear program (integer-programming pricing), or price on the convex-hull relaxation where $u_{jt} \in [0, 1]$ instead of $u_{jt} \in \{0, 1\}$. We adopt this convention throughout: in the proofs and in the decomposition of Proposition 2, λ_{nt} and $\pi_{\ell t} \geq 0$ denote the linear-program multipliers of this convex restriction, the per-zone locational prices and corridor shadow prices respectively. The gap between the mixed-integer optimum and its convex-hull relaxation is itself a nonconvexity term, which we report separately rather than fold into the congestion rent (Remark 3). This is the nodal benchmark, on top of which we compute the make-whole-minimising PE-A prices of Bichler et al. [5] and their total MWP^{PEA} .

3.3. Stage 1: network-blind national clearing

GB clears energy without the network, collapsing the zonal balances of Problem (2) into a single per-period system balance, and dropping the corridors and their flows, gives

$$\begin{aligned} \tilde{W} = \max_{x, y, u, d} \quad & \sum_{i,t} v_{it} x_{it} - \sum_{j,t} (c_{jt} y_{jt} + h_{jt} u_{jt}) \\ \text{s.t.} \quad & \sum_n p_{nt} = 0 \quad \forall t \quad (\lambda_t^{\text{nat}}) \\ & (y, u, d) \in \mathcal{T}_j, \quad 0 \leq x_{it} \leq \bar{D}_{it}, \quad u_{jt} \in \{0, 1\}. \end{aligned} \quad (3)$$

Because Problem (3) relaxes Problem (2) (the corridor constraints are removed), $\tilde{W} \geq W^*$. The single dual λ_t^{nat} of the system balance is the *national price*: one anonymous price per settlement period, common to all zones. The optimiser $(\tilde{x}, \tilde{y}, \tilde{u})$ is the self-schedule, which is generically infeasible for Problem (2): substituting $\tilde{p}_{nt}$ into the corridor definition yields flows that violate some $\bar{B}_{\ell t}$. Because λ^{nat} comes from a mixed-integer program, in computation we recover it as the dual of the balance in the linear program obtained by fixing $u = \tilde{u}$, this is the integer-programming pricing approach of O'Neill et al. [30]. Operationally this stage is decentralised – bilateral, day-ahead and intraday trading – rather than a central solver; we use Problem (3) as its competitive proxy and, empirically, replace $\tilde{y}$ by the observed self-schedule (§3.5).

3.4. Stage 2: redispatch in the Balancing Mechanism

To make the self-schedule deliverable, NESO accepts turn-up offers (price o_{jt}) and turn-down bids (price b_{jt}), moving generation by $\Delta y_{jt} = y_{jt} - \tilde{y}_{jt}$. It chooses the network-feasible dispatch of least redispatch cost:

$$\begin{aligned}
 RC(o, b) = \min_{x, y, u, d, f} \sum_{j, t} & \left[o_{jt}(y_{jt} - \tilde{y}_{jt})^+ - b_{jt}(y_{jt} - \tilde{y}_{jt})^- \right] \\
 \text{s.t. } p_{nt} = \sum_{\ell} A_{n\ell} f_{\ell t} & \quad \forall n, t \quad (\lambda_{nt}) \\
 -\bar{B}_{\ell t} \leq f_{\ell t} \leq \bar{B}_{\ell t} & \quad \forall \ell, t \quad (\pi_{\ell t}) \\
 (y, u, d) \in \mathcal{T}_j, \quad 0 \leq x_{it} \leq \bar{D}_{it}, \quad u = \tilde{u}, &
 \end{aligned} \tag{4}$$

with $(z)^+ = \max(z, 0)$ and $(z)^- = \max(-z, 0)$. The redispatch multipliers $\lambda_{nt}, \pi_{\ell t}$ carry the same symbols as the nodal benchmark's in (2) by design: with cost-reflective bids ($o = b = c$), Proposition 1 shows the redispatch reproduces the benchmark dispatch, so at the optimum the two programs' duals coincide – both are the convex-restriction multipliers fixed once in §3.2.

Three important remarks

- (i) *Commitment.* Problem (4) holds self-commitment fixed at $\tilde{u}$ (the “tight-BM” case): the redispatch may move *committed* units up and down but may not switch units on or off. The alternative, in which the balancing stage is allowed to start additional units by paying their start-up and no-load costs, relaxes the last constraint to $u_{jt} \geq \tilde{u}_{jt}$ and is what we call *recommitment*. The two are exposed as a switch in the solver (`commitment_policy`), and we show in Remark 1 below (after Proposition 1) that the choice between them moves the inefficiency between a welfare loss and an out-of-market side payment rather than eliminating it; the tight-BM case is the one that makes Proposition 1(b) sharpest.
- (ii) *Strategic markup.* Cost-reflective redispatch sets $o = b = c$; strategic conduct is captured by a markup $o_{jt} = c_{jt}(1 + \mu_{jt})$, $\mu_{jt} \geq 0$, which we *estimate* from the data (§5) – finding it indistinguishable from zero at B6 – rather than assume, and whose effect we trace as a counterfactual in §6.4.
- (iii) *Curtailment of subsidised wind.* A support-backed wind unit requires compensation to reduce output, so its turn-down bid is negative, $b_{jt} < 0$; the term $-b_{jt}(\Delta y_{jt})^-$ then contributes a positive cost, which is precisely the curtailment-payment channel that dominates GB constraint costs. Price-sensitive demand enters symmetrically through buyer-side acceptances valued at v . The realised physical welfare is $W^{\text{BM}} = W(x^{\text{BM}}, y^{\text{BM}}, u^{\text{BM}})$, and total consumer cost is the national energy payment at λ^{nat} plus the socialised redispatch cost RC , recovered from demand through the Balancing Services Use of System (BSUoS) charge rather than attributed to identified units.

3.5. From data to model: the settlement correspondence

The two stages are not a stylisation imposed on GB, they are its settlement architecture, and every object in the model is a named, publicly reported settlement record (Figure 3), each published by Elexon through the Balancing

Mechanism Reporting Service [12]. Concretely:

- The Stage-1 self-schedule $\tilde{y}$ is the *Final Physical Notification* (FPN): the half-hourly MW profile each Balancing Mechanism Unit declares at gate closure, one hour ahead of real time. This is the realised, network-blind self-dispatch.
- The Stage-2 prices (o, b) are the *Bid–Offer Data* (BOD): the price–volume ladders each unit submits, an *offer* price to turn up and a *bid* price to turn down (the latter negative for support-backed wind that must be paid to curtail).
- The accepted redispatch Δy is read from the *Bid–Offer Acceptance Level Flagged* dataset (BOALF), which records each acceptance as a new instructed MW level. Subtracting the FPN gives the accepted volume. Each acceptance carries a *system (SO) flag*; the SO-flagged acceptances are those NESO takes for system – i.e. transmission-constraint – reasons rather than for energy balancing, and are exactly the redispatch Δy of our Stage 2.

Returning to Example 1: the 800 MW Scottish-wind FPN, its negative turn-down bid in BOD, and the SO-flagged BOALF acceptance that curtails it by 200 MW (with the matching English-gas turn-up) are precisely the records this correspondence maps onto $\tilde{y}$, b , and Δy . Consequently the redispatch cost RC – and, as shown later, its decomposition in Proposition 2 – is largely measurable directly from settlement data, which we exploit both to calibrate the model and to validate its output against NESO’s published constraint cost.

3.6. National pricing in the design framework

Bichler et al. [5] organise market designs by which of the desiderata they retain: linear and anonymous prices (LA), individual rationality (IR), budget balance (BB), and envy-freeness (EF), with side payments restoring IR where prices alone cannot. National-pricing-with-redispatch is the extreme point of this framework at which *locational* envy-freeness is relaxed entirely: a single anonymous price treats a constrained Scottish generator and an English one identically, and physical feasibility is then bought back through personalised, pay-as-bid redispatch. Table 1 places the three regimes side by side; the progression from nodal to national trades transparency of the side payment (attributed make-whole versus socialised redispatch) for simplicity and a single public price.

4. Theoretical results

Preliminaries. Given a feasible solution (x, y, u) , its welfare is $W(x, y, u) = v'x - c'y - h'u$. We compare two feasible sets, both at fixed demand where x is inelastic and price-sensitive otherwise. The *network-feasible* set $\mathcal{F}$ collects all (x, y, u, d, f) satisfying the network-constrained dispatch constraints of the nodal benchmark (2) – the boundary-transfer limits $|\sum_{\ell \in \beta} f_{\ell t}| \leq \bar{B}_{\beta t}$ together with the generators’ technical sets; its maximiser is (x^*, y^*, u^*) with value W^* . The *network-blind* set $\tilde{\mathcal{F}}$ drops the flow-definition and boundary-limit constraints; its maximiser is the self-schedule

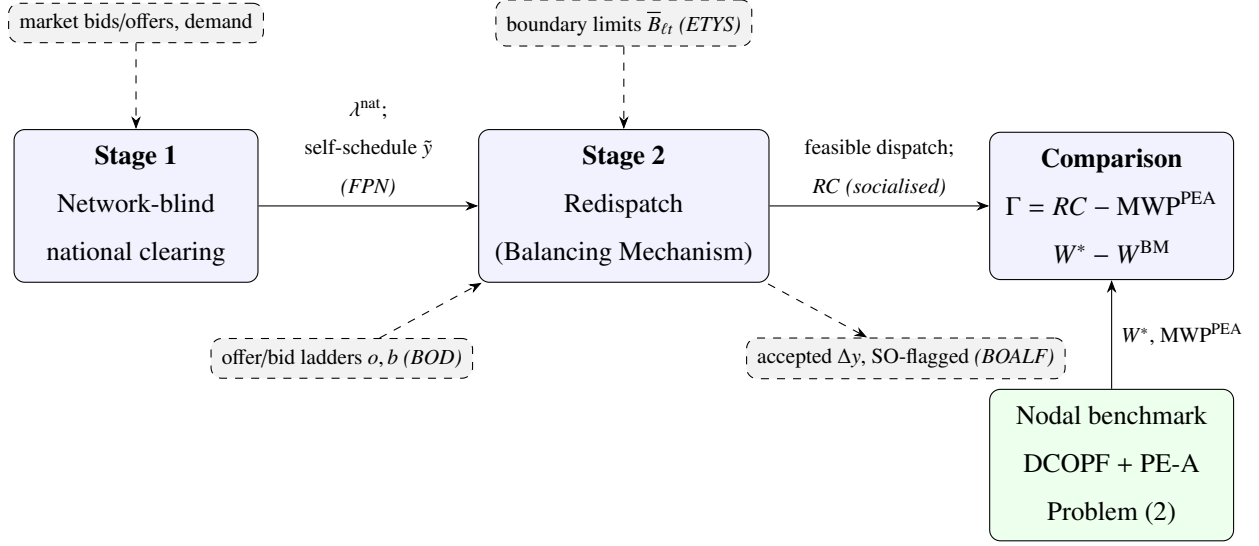


Figure 3: The two-stage model and its mapping to GB settlement data. A network-blind national clearing fixes the single price λ^{nat} and the self-schedule $\tilde{y}$ (observed as FPNs); redispatch then restores network feasibility using boundary limits (ETYS) and the submitted bid-offer ladders (BOD), with the accepted, system-flagged volumes recorded in BOALF. The nodal benchmark supplies W^* and the minimised make-whole payments against which the cost of national pricing Γ is measured.

Table 1: Three regimes in the design framework of Bichler et al. [5]. “Locational EF” is envy-freeness across locations; side payments restore IR in all three.

Regime	Public prices	Locational EF	BB	Side payments
Nodal (PE-A)	per node	relaxed (2nd order)	near	small, attributed (MWP)
Zonal	per zone	partial	no	redispatch (intra-zone)
National + redispatch	single	none	no	redispatch, socialised (BSUoS)

$(\tilde{x}, \tilde{y}, \tilde{u})$ with value $\tilde{W}$. Dropping constraints enlarges the feasible set, so $\tilde{W} \geq W^*$. Given the Stage-1 quantities and the offer/bid prices o, b , the redispatch cost $RC(o, b)$ and the realised dispatch W^{BM} are those of (4). We state each result with its economic interpretation here and delegate all the formal proofs to Appendix A.

Proposition 1 (Nonconvexity is the sole source of inefficiency). *Assume cost-reflective redispatch, $o = b = c$.*

(i) *If the economy is convex (no-load costs $h = 0$, the commitment binaries relaxed, and output is continuously divisible), then $W^{\text{BM}} = W^*$: national-pricing-with-redispatch is welfare-equivalent to nodal pricing.*

(ii) *If commitment u is fixed at $\tilde{u}$, then $W^{\text{BM}} \leq W^*$, and $W^* - W^{\text{BM}} = 0$ only if $\tilde{u} = u^*$.*

The economic reading is that national-pricing-with-redispatch is not intrinsically inefficient; in a convex world it reproduces the nodal optimum exactly, because cost-reflective redispatch merely re-solves the dispatch problem subject to the network. Any realised welfare loss is therefore purely driven by nonconvexities: because Stage 1 is

blind to the network, it may commit the wrong set of units (switch on a generator the network cannot accommodate, or leave off one it needs), and with commitment frozen at $\tilde{u}$ the redispatch stage can re-dispatch within that committed fleet but cannot start or stop units to repair the commitment error.

Part (i) on its own restates a property that is well understood in the locational-pricing literature: with cost-reflective bids a redispatch that restores network feasibility merely re-solves the dispatch program, so in a convex economy it lands on the nodal optimum. We state it not for novelty but because it does two things for the rest of the paper. First, it pins every unit of realised welfare loss to commitment nonconvexity (part (ii)): in the convex world there is none, so whatever loss is observed must come from the fixed commitment $\tilde{u}$. Second, it tells us how to read the decomposition of Proposition 2: because the convex benchmark carries *no* efficiency loss, the congestion-rent term R_{cong} is a pure *transfer* – money moved from consumers to generators – rather than a resource cost. The contribution is that pairing, and its measurement on settlement data, not the convex equivalence on its own.

Remark 1 (The welfare loss is bounded by the BM’s commitment freedom). If the BM may re-optimize commitment – standing self-committed units down where the network cannot use them, and starting those it needs – the constraint $u = \tilde{u}$ is relaxed and the feasible set expands toward $\mathcal{F}$, so the welfare loss $W^* - W^{\text{BM}}$ of Proposition 1(b) shrinks, reaching zero under full recommitment. However, the recovery is not free: it is achieved through out-of-market commitment actions (start-up/no-load payments, or standing units down), settled outside the energy price rather than through it. The realised loss therefore lies between zero (full recommitment) and the tight-BM value, with the position governed by how much recommitment the BM performs (the `commitment_policy` switch in the experiments); the tight-BM case ($u = \tilde{u}$) is the conservative upper bound and the one for which Proposition 1(b) is sharpest. Either way the inefficiency does not vanish – it is relocated from a visible welfare loss to the out-of-market commitment layer.

We now decompose the visible cost. Write $\Gamma = RC - \text{MWP}^{\text{PEA}}$ for the gap between the socialised redispatch cost under national pricing and the minimised make-whole payments MWP^{PEA} that nodal PE-A pricing [5] would instead incur. As in the nodal benchmark (§3.2), the prices $\pi_{\beta t} \geq 0$ below are the boundary-limit multipliers of the convex (LP/convex-hull) restriction of the dispatch.

Proposition 2 (Side-payment decomposition). *Let y^* be the solution for Problem 2, and $\tilde{y}$ the self-schedule for Problem 3. Then with cost-reflective prices augmented by a turn-up markup $o_{jt} = c_{jt}(1 + \mu_{jt})$, $\mu_{jt} \geq 0$ we have the following:*

- (i) (Non-negativity) $RC = c'y^* - c'\tilde{y} \geq 0$, with equality iff the self-schedule is network-feasible.
- (ii) (Decomposition) $RC = R_{\text{cong}} + M$, where the congestion rent $R_{\text{cong}} = \sum_{\beta,t} \pi_{\beta t} s_{\beta t}$ is the boundary shadow price times the relieved overload $s_{\beta t}$, and the strategic markup $M = \sum_{j,t} \mu_{jt} c_{jt} (\Delta y_{jt})^+$ is the surcharge on accepted turn-ups.

Remark 2 (The side-payment gap). The cost of national pricing relative to nodal pricing is the gap $\Gamma = RC - \text{MWP}^{\text{PEA}} = R_{\text{cong}} + M - \text{MWP}^{\text{PEA}}$ (from (ii)), which is non-negative whenever $\text{MWP}^{\text{PEA}} \leq R_{\text{cong}} + M$. This condition

is mild: Bichler et al. [5] report that PE-A make-whole payments are negligible in their experiments (the very feature that makes PE-A attractive), so the right-hand side ($R_{\text{cong}} + M$), which contains the full congestion rent, dominates the left (MWP^{PEA}) whenever a boundary binds. The economic content is that the congestion rent nodal pricing would *collect* as merchandising surplus (and could rebate to consumers) is, under national pricing, instead *paid out* as redispatch and socialised; to first order $\Gamma \approx R_{\text{cong}} + M$, so on this margin national pricing is never cheaper than nodal. The non-negativity is a binding-boundary statement: where no boundary binds, $RC = R_{\text{cong}} = M = 0$ and $\Gamma = -\text{MWP}^{\text{PEA}} \leq 0$ – national pricing is then (weakly) cheaper, precisely because it avoids the nodal make-whole – so the cost of national pricing arises only when the network constrains the self-schedule.

Remark 3 (Nonconvex shadow prices). With binary commitment the dispatch is a mixed-integer program and exact dual prices $\pi_{\beta t}$ are not defined, so Proposition 2(ii) is taken on the LP/convex-hull relaxation of the dispatch (binaries relaxed, or convex-hull prices in the sense of [14]) – the same object NESO-style locational pricing would use. The gap between the mixed-integer redispatch cost and its convex-hull relaxation is itself a nonconvexity term; we report it separately rather than fold it into R_{cong} .

4.1. Optimal siting of flexibility across nested boundaries

Proposition 2 values a *given* placement of flexibility behind a single cut, which raises the question: given a budget of flexible capacity – storage or shiftable demand – *where* along the network should it go, and how should it be operated over the day, to reduce the cost of national pricing the most? With several *nested* GB boundaries (e.g., B4 north of B6) and storage that moves energy across time, this is a genuine optimisation, which we now state and solve.

Order the zones along the radial cascade $z = 1, \dots, Z$ from the surplus (northern) end to demand, with nested cuts $\beta = 1, \dots, B$; cut β separates the first z_β zones from the rest, $z_1 < \dots < z_B$ (see Figure 2), so energy injected at zone z must cross every cut β with $z_\beta \geq z$ to reach demand. A budget K of flexible storage power is sited, $k_z \geq 0$ with $\sum_z k_z \leq K$; each store has energy $k_z D$ (duration D) and round-trip efficiency η , with charge and discharge bounded by k_z and the usual state-of-charge dynamics across periods t . Writing $C(K)$ for the minimised socialised redispatch (equivalently generation) cost when the network limits are enforced, and $V(K) = C(0) - C(K)$ for the value of the budget, the siting-and-operation program is

$$\begin{aligned}
C(K) = \min_{k, g, w, F, \sigma, ch, dis} \quad & \sum_{z, t} c_z g_{zt} \\
\text{s.t.} \quad & w_{zt} + g_{zt} + dis_{zt} - ch_{zt} - x_{zt} = \sum_{\beta} A_{z\beta} f_{\beta t} & \forall z, t \quad (\lambda_{zt}) \\
& -\bar{B}_{\beta t} \leq f_{\beta t} \leq \bar{B}_{\beta t} & \forall \beta, t \quad (\pi_{\beta t}) \\
& \sigma_{zt} = \sigma_{z, t-1} + \sqrt{\eta} ch_{zt} - dis_{zt} / \sqrt{\eta} & \forall z, t \\
& ch_{zt} \leq k_z, \quad dis_{zt} \leq k_z, \quad \sigma_{zt} \leq k_z D & \forall z, t \quad (\rho_{zt}) \\
& \sum_z k_z \leq K, \quad k_z \geq 0, \quad 0 \leq w_{zt} \leq \bar{W}_{zt}, \quad g_{zt} \geq 0. & (\mu)
\end{aligned} \tag{5}$$

Here g_{zt} is dispatchable (replacement) generation at cost c_z , $w_{zt} \leq \bar{W}_{zt}$ the used wind, x_{zt} the inelastic demand, $f_{\beta t}$ the flow on cut β (incidence $A_{z\beta}$), and $(\sigma_{zt}, ch_{zt}, dis_{zt})$ the state of charge, charge and discharge of a store in zone z , all capped by the sited power k_z (energy $k_z D$, round-trip η). The parenthesised symbols are the constraint duals: λ_{zt} (zonal balance), $\pi_{\beta t} \geq 0$ (the cut limits – the congestion prices of Proposition 2), $\rho_{zt} \geq 0$ (the store’s power/energy caps) and $\mu \geq 0$ (the siting budget). Write $\lambda_z := \partial V / \partial k_z \geq 0$ for the per-MW marginal value of capacity in z ; by LP sensitivity it is the aggregate shadow value of zone z ’s capacity caps. Problem (5) is solved by `flex_siting.py`.

Proposition 3 (Marginal value and water-filling of flexibility). *At an optimum of (5):*

- (i) *the marginal value $\lambda_z = \partial V / \partial k_z$ equals the day-value a store in z extracts by relieving the cuts on its path to demand; by LP duality it is the optimal arbitrage of the shadow prices $\pi_{\beta t}$ of those binding cuts, $\lambda_z \geq 0$.*
- (ii) (Cascade monotonicity.) *Because the cuts are nested, a store further upstream relieves a superset of the cuts a downstream store does, so λ_z is non-decreasing up the cascade.*
- (iii) (Water-filling.) *The budget KKT condition is $\lambda_z \leq \mu$ with equality where $k_z > 0$: the optimal siting fills the highest-marginal-value (most upstream binding) zone first and moves down the cascade, an interior split equalising marginal values across the zones used. The single-cut, single-period case recovers the flexibility-equivalence of §7, $\lambda = \pi_\beta$.*

5. Computational design

Models. The computation mirrors the model’s structure. Because GB national pricing is not one optimisation but two solved in sequence – a network-blind clearing followed by a redispatch that takes the clearing’s output as fixed data – we implement the three programs of §3 separately:

1. the network-blind national clearing of Stage 1 Problem (3), which determines the self-schedule $\tilde{y}$ and the national price;
2. the cost-minimising redispatch of Stage 2 Problem (4), with two switches – cost-reflective ($o = b = c$) versus marked-up offers, and commitment held fixed (tight BM) versus allowed to recommit;
3. the nodal PE-A make-whole minimisation of Bichler et al. [5], adapted to the boundary-transfer network, which supplies the benchmark MWP^{PEA} .

All three are mixed-integer/linear programs over the same boundary-transfer network, and from the Stage-2 boundary duals we read off the decomposition $RC = R_{\text{cong}} + M$ of Proposition 2.

Test systems. We use two test systems of increasing data realism, so that the mechanism can first be checked against known costs and then measured on real data. The *first* (controlled) setting is a small *purpose-built* two-zone unit-commitment instance with known costs (`synthetic_b6` and `synthetic_b6_nonconvex`): a convex version that

isolates the congestion rent and a nonconvex version – committable generators with no-load and minimum-load on both sides of B6 – in which the network-blind clearing commits the wrong units, so the comparative statics of Propositions 1–2 can be read off against ground truth (§6.1). The *second* (empirical) setting, which supplies external validity, is the reduced GB B6 model of §5, with boundary capabilities from ETYS (and the observed, outage-adjusted SCOTEX limit), generation anchored to FUELHH and demand to the national outturn, and Balancing-Mechanism market data from Elexon.

The settlement–model correspondence. The two stages map exactly onto observable GB settlement objects: the Stage-1 self-schedule is the Final Physical Notification (FPN) at gate closure; the Stage-2 redispatch is the set of bid-offer acceptances carrying the system (SO) flag – the flag NESO attaches to an acceptance taken to manage a transmission constraint, as distinct from the unflagged acceptances used to balance national energy supply and demand – priced at the submitted Bid-Order Data ladders. The decomposition of Proposition 2 is therefore largely observable.

Calibration, identification and validation. Generators’ no-load and start-up costs are not public; we assign them by technology archetype (e.g., wind, gas) and calibrate them so that the model reproduces NESO’s published aggregate transmission-constraint cost for the day (the figure NESO reports in its Balancing Costs publications, $\approx$ £1.9bn for 2024/25 at the annual level), reporting Proposition 1’s welfare loss in bands rather than as a point estimate. The strategic markup μ is *estimated* from the data rather than assumed, by a regression-discontinuity design that compares the price of system-flagged turn-up acceptances in periods when B6 binds against periods when it does not. The clean specification (`estimate_markup_pooled`) is *within-unit-day*: for a unit turned up in both a binding and a non-binding period of the same day, we measure the proportional change in its *own* turn-up offer, so unit and day fixed effects net out the unit’s marginal cost and the day’s wholesale level – exactly the confounders that defeat a naive offer-versus-wholesale comparison (which is badly upward-biased because binding periods are low-wholesale overnight/high-wind hours). Pooled over the sampled days (83 unit-day pairs in which a unit appears on both sides of the threshold), the estimate is essentially zero: $\hat{\mu} = 0.00$ (s.e. 0.007; 95% CI $[-0.01, 0.02]$), against a spurious +3.4 from the naive cross-unit premium-over-wholesale contrast – the same binding-vs-non-binding comparison *without* the unit and day fixed effects, computed alongside the headline by the same routine (`estimate_markup_pooled`) and reported here only to show the bias the within-unit-day design removes. Identification rests on the demanding subsample of unit-days for which the same unit is accepted in both a binding and a non-binding period, so the count is modest; but the tightness of the interval shows this is a precisely estimated zero rather than an underpowered null – it rules out any boundary markup above about two percent of the wholesale price, far below the $\sim 30\%$ replacement-gas premium documented in §1 that motivates the markup channel. Units do not raise their turn-up offers when B6 binds: the visible cost is congestion rent, not a strategic markup – consistent with the inc-dec literature’s reading that redispatch cost reflects market *design* rather than market power [16]. We therefore take the cost-reflective case ($\mu = 0$) as the data-supported headline and treat a positive markup as a counterfactual (the μ -sweep of §6.4).

We reconcile a bottom-up redispatch cost computed directly from SO-flagged acceptances – each instructed level held over its half-hour settlement period and priced at the submitted Bid-Offer Data, decomposed into Scottish-wind curtailment and replacement turn-up – against NESO’s published constraint cost for the same day. Two points of scope matter. The bottom-up figure is the cost of *Balancing Mechanism* bid-offer acceptances, which is the dominant component of, but smaller than, NESO’s published *system* constraint cost: the latter additionally includes constraint-management actions taken outside the BM (constraint trades, interconnector actions and other balancing services) that are not recorded in the BOALF dataset. We therefore validate that the BM bid-offer cost is a large, plausible share of – without exceeding – the published total (empirically $\approx 79\%$ on the selected day), rather than demanding equality, and quote the headline only when this holds.

6. Numerical and empirical results

We report the two settings in turn: a controlled nonconvex instance that exhibits the welfare-loss result of Proposition 1 against known costs (§6.1), then the GB B6 boundary (§6.2 onward).

6.1. A controlled nonconvex instance (Proposition 1(b))

In the GB aggregate the realised welfare loss is essentially zero (§6.2), as Proposition 1(a) predicts for a near-convex economy: the cost of national pricing there is congestion rent, not lost dispatch efficiency. To exhibit the welfare loss of part (b) – the genuinely nonconvex case – we use the purpose-built instance of §5 with known costs (`synthetic_b6_nonconvex`; full walk-through in the code). Demand is 100 (SCO) + 1000 (ENG) MW across a 500 MW B6 cut, with committable units carrying no-load and minimum-load on both sides.

The network-blind clearing commits a cheap committable Scottish unit ($\tilde{u} = \{\text{SCO_CCGT}, \text{ENG_BASE}\}$; the costs are illustrative, not technology-calibrated), but the network optimum serves the north with wind alone and leaves that unit off ($u^* = \{\text{ENG_BASE}\}$): $\tilde{u} \neq u^*$. Under the tight BM the mis-committed unit cannot be stood down – it is stranded at its minimum stable load behind B6 – so $W^{\text{BM}} < W^*$. Table 2 reports the result: a welfare loss of £4,800, decomposing into a £2,800 minimum-load component (the stranded unit displacing cheaper wind) and a £2,000 no-load component, and growing linearly with the no-load cost. The loss is exactly zero in the convex instance, confirming Proposition 1(a), and is recovered in full by recommitment, as in Remark 1.

6.2. The GB B6 boundary: a peak-curtailment day

We select the peak-curtailment day in winter 2024/25 by ranking candidate days on total system-flagged Scottish-wind turn-down volume, then solve every settlement period and aggregate. Table 3 reports the daily totals; the reconciliation against the published constraint cost is reported alongside.

The validation panel confirms the data pipeline: the bottom-up cost of SO-flagged BM bid-offer acceptances – every flagged acceptance on the day priced at its submitted Bid-Offer Data and summed over the 48 periods

Table 2: Proposition 1 on controlled instances with known costs (`nonconvex_experiment.py`). The welfare loss is a purely driven by commitment-nonconvexity: zero when convex, growing with the no-load cost, recovered by recommitment.

Instance / regime	Welfare loss $W^* - W^{\text{BM}}$ (£)
Convex (<code>synthetic_b6</code>)	0
Nonconvex, tight BM	4,800
of which minimum-load component (no-load $\leftarrow 0$)	2,800
of which no-load component	2,000
Nonconvex, recommitment	0
<i>Comparative static: tight-BM loss vs no-load scale θ</i>	
$\theta = 0$ (min-load only)	2,800
$\theta = 0.5$	3,800
$\theta = 1.0$ (baseline)	4,800

(`redispatch_cost_detailed`), £8.76m – is 79% of NESO’s published system constraint cost (£11.1m), the residual being the non-BM constraint actions discussed in §5. This day-level figure is the reconciliation anchor (it checks that our reading of the settlement data is sound); it is not the headline, which is the directly-measured annual rent of §6.3. The reconciliation passes, so the modelled figures are quoted. The modelled quantities are the B6 two-zone reduction under the central calibration; the welfare loss is zero because the aggregate is convex, so – exactly as in the worked Example 1, where $RC = R_{\text{cong}} = £11,600$ and $MWP^{\text{PEA}} = 0$ – the entire cost of national pricing surfaces as $\Gamma = R_{\text{cong}}$, here $\approx £2.88\text{m}$ for the day. The markup counterfactual ($\mu = 0.30$, the \$1 replacement-gas premium) adds £0.37m, but the data-driven markup is ≈ 0 (§5), so the cost-reflective column is the one we carry forward.

Consistent with Proposition 1(a), in the aggregated convex representation the realised welfare loss is zero and the cost of national pricing surfaces entirely as the congestion rent R_{cong} (the markup $M \approx 0$ in the data); the data validation closes the loop: reconstructing the SO-flagged BM constraint cost from settlement records recovers 79% of NESO’s published system constraint cost, the remainder being constraint actions taken outside the Balancing Mechanism, which confirms that our reading of the settlement data is sound.

6.3. From a day to a year: a data-anchored annual cost

A single day cannot speak to the magnitude that drives the policy debate – NESO’s annual constraint cost of $\approx £1.9\text{bn}$ – so we scale to a year, but doing so exposes a calibration question that we resolve with observed boundary data.

Why the structural model is not used for the level. The two-zone instance of §5 reproduces the *mechanism* of Proposition 2 and is well suited to the comparative statics below, where only *relative* responses matter. Its absolute level,

Table 3: Modelled cost of national pricing on the B6 boundary, 8 December 2024 (daily totals, £; FUELHH/demand-anchored two-zone instance, central calibration; `run_paper.py`). The cost-reflective column is the data-supported case ($\hat{\mu} \approx 0$; §5); the second is a $\mu = 0.30$ *counterfactual* (set to the $\approx 30\%$ replacement-gas premium over wholesale documented in §1). The rows read top-to-bottom as the Proposition 2 decomposition. The data validation is in the text.

Quantity	Cost-reflective ($o = b = c$)	Markup counterfactual ($\mu = 0.30$)
Congestion rent R_{cong}	2,883,000	2,883,000
+ strategic markup M	0	368,000
= redispatch cost RC	2,883,000	3,251,000
– nodal make-whole MWP^{PEA}	0	0
= gap $\Gamma = RC - MWP^{\text{PEA}}$	2,883,000	3,251,000
Welfare loss $W^* - W^{\text{BM}}$ (Prop. 1)	0	0

however, is sensitive to two stylised inputs – how much GB wind sits behind B6, and the boundary capability. Anchoring the capability to NESO’s published, outage-adjusted B6 (SCOTEX) limit, which falls as low as 2.5 GW on fault days rather than the ≈ 6 GW ETYS headline, swings the modelled annual rent from £165m (fixed limit) to £754m (observed fault-day limits, ± 142 SE) – both below the directly measured £1.1bn of the following section. The structural reduction cannot, with a single Scottish-wind share, simultaneously match the observed curtailment volume and the days on which it occurs, because the effective capability varies day-to-day with network faults that a fixed two-zone reduction does not represent, so it understates the realised cost. We therefore do not quote the structural model as the level; we measure the level directly.

A realised, settlement-data measure. The relieved overload is *observed*: it is the volume of system-flagged Scottish-wind turn-down in BOALF, i.e. the curtailment NESO actually instructed to keep B6 within its limit. Scottish (north-of-B6) units are identified from the BMRS registry by GSP group together with the zonal transmission loss factor (the latter is essential, as transmission-connected units carry no GSP group); Appendix C shows the annual figure is invariant to the loss-factor cut and cross-checks the selected units by name. Pricing each curtailed MWh at the boundary rent – the wholesale replacement price south of the cut less the unit’s (negative) turn-down bid – gives the realised B6 congestion rent directly from settlement records, with no zonal-share assumption. (we use the SCOTEX *limit* but not its *flow* field, which is a day-ahead *unconstrained* forecast that can exceed B6’s physical capacity and so overstates the realised overload.)

Annualisation. Daily costs are highly skewed (the busiest tenth of days carries about a third of the annual total), so we estimate the annual total by *stratified sampling*, using NESO’s published daily series as both the stratifying variable and an independent anchor. We retrieve NESO’s daily *system* constraint cost for all 365 days of FY 2024/25 (it sums

to £1,903m), sort the days into five equal-count bins by that cost, draw a random sample of six days per bin (30 days), compute the realised B6 rent for each, and form the stratified estimator $\widehat{C} = \sum_b N_b \bar{c}_b$ with its standard error (N_b the days in bin b , $\bar{c}_b$ the sample mean there). This is `run_annual.py`.

Table 4 reports the result. The realised B6 congestion rent tracks the independent NESO daily series almost exactly (day-level correlation 0.97) and aggregates to £1,126m/yr ($\pm$ £98m), on 9.55 TWh of Balancing-Mechanism-visible Scottish-wind curtailment – about 59% of the system-wide constraint cost. The comparison is like-for-like: both the realised rent and NESO’s published constraint cost are the payment to turn Scottish wind *down* plus the payment to turn replacement generation *up*, so the share is a genuine fraction of the same quantity rather than a ratio of differently-defined costs – and if anything it is conservative, because we value the replacement leg at the wholesale index rather than at the (higher) accepted offer price NESO actually pays. National-pricing redispatch on the single B6 boundary therefore costs of order £1bn/yr and is the dominant component of GB constraint costs.

Two checks support the magnitude. *External corroboration*: our settlement-derived 9.55 TWh (FY 2024/25) is consistent with independently published curtailment volumes – the Renewable Energy Foundation, also working from BMRS settlement records, reports 8.3 TWh of GB wind curtailed in calendar 2024, the great majority Scottish [31], and FY 2024/25 extends into the higher-curtailment 2025 period – so the figure matches the public record rather than being a partial measure. *B6 attribution*: 97% of the curtailed volume falls in settlement periods where the real-time (physical-notification-implied) northward flow exceeds NESO’s outage-adjusted SCOTEX limit, confirming the turn-down relieves the Scotland–England constraint rather than local intra-Scotland faults (the day-ahead SCOTEX *flow* field is an unconstrained forecast and under-predicts on exactly the high-wind days, so it is not used for this test). The 41% residual of the system total reflects other boundaries, thermal and voltage actions, and non-curtailment constraint costs. The one component our Balancing-Mechanism measure still omits is curtailment of *embedded* (distribution-connected) Scottish wind, which settles outside the Mechanism; including it would raise the figure modestly further, so £1.1bn is, if anything, a mild lower bound on the full B6 cost. A complete embedded-inclusive figure requires the multi-zone reduced-network build, which is left to future work.

6.4. Scaling and sensitivity

Because the policy interest is in how the cost of national pricing moves with the levers REMA can pull, we sweep each of four levers (B6 capability, Wind penetration, Flexibility behind B6, and Strategic markup) on the same day, holding the others at their central calibration, with `sensitivity.py`. Each sweep re-solves all 48 periods, Figure 4 reports the resulting daily totals, and the full grids are in Appendix B.

The four statics are exactly the policy levers discussed in §7, and each behaves as the theory predicts.

- *B6 capability*. Transmission build-out reduces the relieved overload one-for-one: widening B6 from 6.1 to 7.0 GW cuts R_{cong} from £2.9m to £0.3m per day, and at ≈ 8 GW the self-schedule is network-feasible and $R_{\text{cong}} = 0$, precisely Proposition 2(i).

Table 4: Annualised cost of national pricing on the B6 boundary, FY 2024/25, from a 30-day stratified sample (`run_annual.py`). The realised rent is measured directly from settlement data; the NESO figure is the published *system-wide* annual constraint cost (all boundaries and causes). “SE” is the standard error of the stratified estimator.

Quantity	Value
<i>Headline (data-driven)</i>	
Realised B6 congestion rent	£1,126m/yr (± 98 SE)
<i>Diagnostics (not components of the rent)</i>	
BM-visible Scottish-wind curtailment	9.55 TWh/yr
Correlation of daily rent with NESO series	0.97
Realised B6 rent as share of system total	59%
Curtailment in real-time B6-binding periods	97%
<i>For reference</i>	
Published GB wind curtailment, CY2024 [31]	8.3 TWh/yr
NESO published <i>system</i> constraint cost	£1,903m/yr

- *Wind penetration (Clean Power 2030)*. Above the binding threshold R_{cong} rises essentially linearly – about £6m per day for each $0.2\times$ increment of Scottish wind output – so the cost of retaining national pricing scales steeply along the FES trajectory; below $\approx 0.9\times$ the boundary stops binding.
- *Flexibility behind the boundary*. Relocating flexible demand or storage north of B6 absorbs the surplus locally: ≈ 2 GW eliminates R_{cong} entirely. This is the flexibility-equivalence result of §7 in numbers – each MW of behind-boundary flexibility is worth the boundary shadow price in avoided socialised redispatch.
- *Markup (counterfactual)*. The strategic component M is linear in μ ($\approx$ £1.2m per unit μ) and additive on the congestion rent, $\Gamma = R_{\text{cong}} + M$; it is the part of the gap that BM-competition measures, rather than transmission or flexibility, would address. Since the within-unit-day estimate is $\hat{\mu} \approx 0$ (§5), panel (d) is a counterfactual: at B6 today $M \approx 0$ and the gap is essentially R_{cong} .

6.5. Optimal flexibility siting (Proposition 3)

Following the analysis from the previous section, we now explore the role of flexibility. We solve the full siting-and-operation program (5) on a stylised nested instance with known costs (`flex_siting.py`): a North Scotland $\rightarrow$ South Scotland $\rightarrow$ England chain across the nested cuts B4 and B6, a diurnal wind profile anti-correlated with the English demand peak, and 4-hour storage of budget K to be sited in the north and/or south. Both cuts bind, so northern storage relieves *both* (it captures wind trapped behind B4 and arbitrages B6) while southern storage relieves *only* B6.

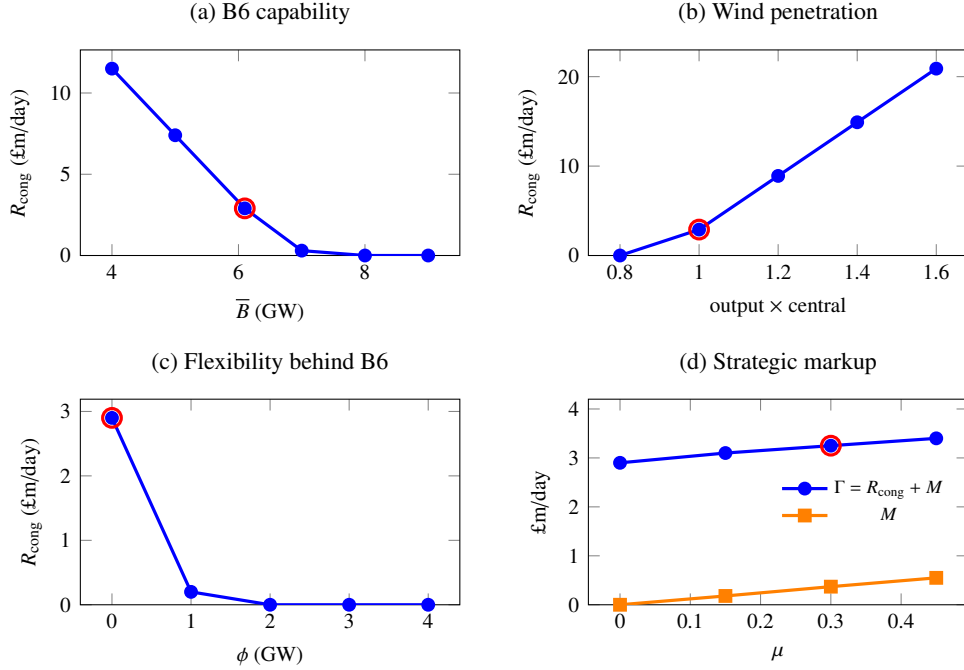


Figure 4: Comparative statics on the B6 boundary, 8 December 2024 (daily totals). Each panel sweeps one lever, holding the others at the central calibration (red ring). Panels (a)–(c) report the congestion rent R_{cong} ; panel (d) the gap $\Gamma = R_{\text{cong}} + M$ and its markup component M . Transmission build-out (a) and behind-boundary flexibility (c) drive the rent to zero; wind growth (b) raises it near-linearly; the markup (d) is additive. Numeric grids in Appendix B; produced by `sensitivity.py`.

Figure 5 reports the value of the budget, $V(K)$, for the optimal siting and for the two pure placements. Three features confirm Proposition 3. First, *siting is decisive*: placing the whole budget in the south captures only $\approx$ £52k/day, against $\approx$ £480k for the north and up to $\approx$ £533k for the optimum – getting the location wrong forfeits roughly nine-tenths of the value. Second, the *optimum water-fills the cascade*: it fills the north (higher marginal value, relieving both cuts) first, then spills into the south once the north’s marginal value has fallen, so the optimal split strictly beats either pure placement once the budget exceeds the north’s useful capacity. Third, the value is *concave and saturates* ($\approx$ £0.53m/day here) once flexibility has absorbed the binding surplus, the multi-period analogue of $R_{\text{cong}} \rightarrow 0$ in §6.4.

7. Discussion and policy implications

The results give a precise reading of REMA’s choice. Retaining national pricing does not, in a near-convex aggregate, sacrifice much dispatch efficiency; its cost is the congestion rent that a locational price would have internalised, here paid out as redispatch and socialised through balancing charges, plus a strategic markup. This reframes the reformed-national-pricing agenda: measures that reduce the binding congestion (transmission build-out, strategic siting) or that place flexible demand and storage behind the boundary act directly on R_{cong} , while measures that improve

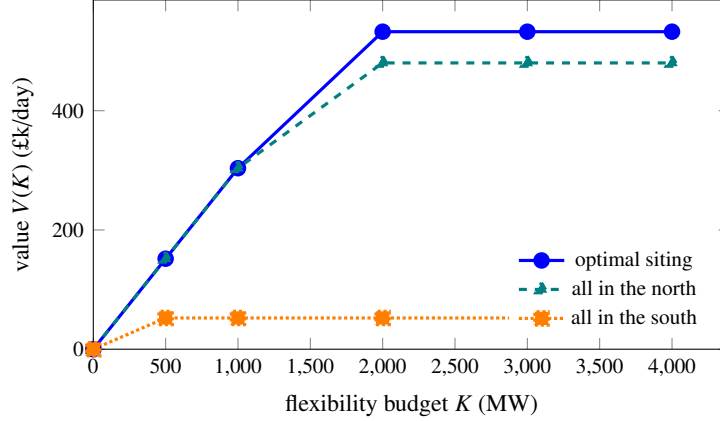


Figure 5: Value of a flexibility budget on the nested B4/B6 cascade (stylised instance, daily totals; `flex_siting.py`). Optimal siting water-fills the north (relieving both cuts) before the south; placing the budget south of the binding B4 cut captures almost none of the value; the value is concave and saturates. This is Proposition 3 in numbers.

BM competition act on M .

We make the flexibility channel precise. Suppose a quantity $\phi_{\beta t} \geq 0$ of flexible resource – price-sensitive demand, storage, or shiftable load – sits north of the binding cut and can absorb surplus locally. Its first-order effect is to reduce the relieved overload one-for-one, $s_{\beta t}(\phi) = (s_{\beta t}(0) - \phi_{\beta t})^+$, so that, by Proposition 2(ii), the congestion-rent component falls as $R_{\text{cong}}(\phi) = \sum_{\beta, t} \pi_{\beta t} (s_{\beta t}(0) - \phi_{\beta t})^+$ and reaches its uplift-only minimum once $\phi_{\beta t} \geq s_{\beta t}(0)$ for all binding (β, t) , i.e. once enough locational flexibility makes the self-schedule network-feasible. This *flexibility-equivalence* result – a given megawatt of behind-the-boundary flexibility is worth the boundary shadow price $\pi_{\beta t}$ in avoided socialised redispatch – formalises the central premise of REMA’s reformed national-pricing model. *Where* that flexibility should sit, when several boundaries bind, is the optimal-siting problem we solve in §4.1: Proposition 3 shows the marginal value of siting rises up the nested cascade (a megawatt in the far north relieves every cut it sits behind, not just the nearest), and that the budget should water-fill the most upstream binding location first. The numerical instance of §6.5 bears this out starkly – siting the budget south of the binding cut forfeits roughly nine-tenths of the value. This turns “strategic siting” from a slogan into a ranked, quantified prescription. The remaining extension, integrating the siting decision with the strategic markup M and the full multi-zone reduced network, is left to future work.

8. Conclusion

This paper has characterised the cost of national electricity pricing as a two-stage phenomenon – a network-blind national clearing followed by a cost-minimising redispatch – and located that institution as the extreme, locational-envy-freeness-relaxing point of the side-payment design framework of Bichler et al. [5]. The contribution is fivefold. First, a model: we formalise GB self-dispatch-with-redispatch as the composition of three programs whose objects coincide exactly with the published settlement records (FPN, BOD, SO-flagged BOALF), so the theory is measurable

rather than merely illustrative. Second, an efficiency result (Proposition 1): national pricing with cost-reflective redispatch is welfare-equivalent to nodal pricing in a convex economy, so any realised welfare loss is a purely driven by commitment-nonconvexity – and, under recommitment, even that loss migrates into an out-of-market side payment rather than vanishing. Third, a decomposition (Proposition 2): the visible cost splits into a congestion-rent term that nodal pricing would *collect* as merchandising surplus but national pricing *pays out* and socialises, plus a strategic redispatch markup, with a gap to the minimised nodal make-whole payments that is non-negative under an empirically supported condition. Fourth, a measurement: exploiting the settlement correspondence and a FUELHH/demand-anchored reduction of the Scotland–England (B6) boundary, we quantify the cost for a peak-curtailement day and scale it, by a settlement-data stratified estimator, to a directly measured annual B6 congestion rent of order £1.1bn in FY 2024/25 ($\approx 59\%$ of GB system constraint cost, correlation 0.97 with the published daily series), validate the bottom-up Balancing Mechanism cost against NESO’s published constraint cost, and trace how the figure responds to the levers in the reform debate – transmission build-out and behind-boundary flexibility both drive the congestion rent to zero, wind growth raises it steeply, and BM competition acts only on the markup. Fifth, an optimisation (Proposition 3): we cast the placement of flexibility across the nested GB boundaries as a siting-and-operation program and show its marginal value rises up the cascade and that the budget should water-fill the most upstream binding location first – a ranked, quantified version of “strategic siting” in which placing flexibility on the wrong side of a binding cut forfeits almost all of its value.

Two readings follow for policy. The efficiency case against national pricing is, in a near-convex aggregate, weaker than often asserted: the first-order cost is not foregone dispatch efficiency but congestion rent that is paid out and socialised rather than internalised and rebated. And the reformed-national-pricing agenda has a precise target – measures that relieve the binding boundary (transmission, siting, storage and flexible demand behind the cut) act directly on R_{cong} , while measures that sharpen balancing-market competition act on M – though at B6 we find M empirically negligible, so the first-order target is R_{cong} . Several extensions are natural: a full multi-zone build on the NESO Reduced Model and Future Energy Scenarios (which would also capture embedded-wind curtailment settled outside the Balancing Mechanism), and joint optimisation of flexibility siting with the strategic markup. More broadly, the framework is not specific to electricity: it applies to any two-stage market that clears at a uniform price and restores feasibility through a subsequent redispatch.

Reproducibility

The optimisation models, including the Proposition 2 decomposition $RC = R_{\text{cong}} + M$ recovered from the boundary LP duals (`gb_two_stage_skeleton.py`), the public BMRS Insights client (`bmr_client.py`), the empirical pipeline wiring PN/BOD/BOALF into the model, estimating the strategic markup μ by the within-unit-day regression discontinuity (`estimate_markup_pooled`) and reconciling the data-derived redispatch cost against the published NESO constraint cost (`gb_empirical_pipeline.py`, function `reconcile`), the orchestration that produces

Table 3 (`run_paper.py`), the controlled Proposition 1 experiment of Table 2 (`nonconvex_experiment.py`), the stratified annualisation of Table 4 (`run_annual.py`), and the flexibility siting-and-operation optimisation of Figure 5 (`flex_siting.py`) accompany this paper. The convex worked Example 1 is reproduced exactly by the bundled `synthetic_b6` instance. All BMRS data are public and require no API key.

CRediT authorship contribution statement

Felipe Maldonado: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The author has nothing to declare.

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Declaration of generative AI in the writing process

During the preparation of this work the author used Claude (Anthropic) as a coding and editing assistant – for code development and debugging of the empirical pipeline, and for text editing. After using this tool, the author reviewed, verified and edited all suggested content and takes full responsibility for the content of the publication.

Data availability

All data are public and require no API key: half-hourly settlement data from the Elexon BMRS Insights API and constraint-cost data from the NESO Data Portal. The code, the cached settlement data, and a README that reproduces every table and figure accompany the paper and are archived at [\[TBD\]](#).

Appendix A. Proofs

Proof of Proposition 1. (i) With $o = b = c$, for any y , the redispatch objective (4) is

$$\sum_j o_j(y_j - \tilde{y}_j)^+ - \sum_j b_j(y_j - \tilde{y}_j)^- = \sum_j c_j(y_j - \tilde{y}_j) = c'y - c'\tilde{y},$$

and $c'\tilde{y}$ is a constant fixed at Stage 1. With demand inelastic, x is fixed and minimising RC over $\mathcal{F}$ is therefore minimising $c'y$ over $\mathcal{F}$, whose minimiser is the efficient dispatch (x^*, y^*) . With demand price-sensitive, the buyer-side terms enter symmetrically and the objective becomes $c'y - v'x - \text{const} = -W(x, y) - \text{const}$; since the economy is convex,

$\mathcal{F}$ is a polytope, so minimising the objective maximises W over $\mathcal{F}$, again at (x^*, y^*) . In either case $W^{\text{BM}} = W^*$. Nodal pricing dispatches the same DCOPF optimum, so the two regimes are welfare-equivalent.

(ii) Fixing $u = \tilde{u}$ confines the redispatch to $\mathcal{F}_{\tilde{u}} = \{(x, y, d, f) : (x, y, \tilde{u}, d, f) \in \mathcal{F}\} \subseteq \mathcal{F}$. By the same algebra the redispatch maximises W over $\mathcal{F}_{\tilde{u}}$, so $W^{\text{BM}} = \max_{\mathcal{F}_{\tilde{u}}} W \leq \max_{\mathcal{F}} W = W^*$, with equality iff the unconstrained-commitment optimum is attained at $u = \tilde{u}$, i.e. iff $\tilde{u} = u^*$. The commitment variable u exists only by virtue of the nonconvex commitment structure, so any loss $W^* - W^{\text{BM}} > 0$ is attributable solely to that nonconvexity. $\square$

Proof of Proposition 2. (i) By Proposition 1, cost-reflective redispatch realises y^* at cost $c'y^* - c'\tilde{y}$. Since $\tilde{y}$ minimises $c'y$ over the larger set $\tilde{\mathcal{F}} \supseteq$ (the projection of) $\mathcal{F}$, we have $c'\tilde{y} \leq c'y^*$, so $RC \geq 0$; equality holds iff $\tilde{y}$ is itself network-feasible, i.e. no boundary binds.

(ii) We argue by complementary slackness: take first a *single* binding boundary β (the empirically relevant B6 case), the self-schedule sends a flow exceeding the capability by $s_{\beta t} = (|\text{flow}_{\beta}(\tilde{y})| - \bar{B}_{\beta t})^+$. To restore feasibility the redispatch must move $s_{\beta t}$ MW of generation from the exporting to the importing side of the cut; at the LP optimum the cheapest way to do so costs, per MW, exactly the difference between the importing- and exporting-side marginal prices, which is the boundary shadow price $\pi_{\beta t}$ (the multiplier on that limit). Hence the cost of relief is $\pi_{\beta t} s_{\beta t}$ and $RC = \pi_{\beta t} s_{\beta t}$. With several boundaries binding, tighten them to their limits one at a time: each step k relieves an increment Δs_k at the prevailing price π_k and adds $\pi_k \Delta s_k$, and the increments adds up to $\sum_{\beta, t} \pi_{\beta t} s_{\beta t} =: R_{\text{cong}}$. Finally, replacing each accepted turn-up offer c_{jt} by $c_{jt}(1 + \mu_{jt})$ adds the surcharge $\sum_{j, t} \mu_{jt} c_{jt} (\Delta y_{jt})^+ = M$ on top, giving $RC = R_{\text{cong}} + M$. $\square$

Proof of Proposition 3. (i) The capacity k_z appears only on the right-hand side of zone z 's power/energy caps (duals $\rho_{zt} \geq 0$), so by LP sensitivity (the envelope theorem) $\partial C / \partial k_z = -\lambda_z$ with $\lambda_z = \sum_t (\rho_{zt}^{ch} + \rho_{zt}^{dis} + D \rho_{zt}^{\sigma}) \geq 0$; hence $\lambda_z = \partial V / \partial k_z \geq 0$. A store in z can shift energy only across the cuts on the path $z \rightarrow$ demand, and by complementary slackness the value of that shifting is the energy arbitrated times the cut prices $\pi_{\beta t}$ over the periods those cuts bind – which is exactly λ_z .

(ii) A store at zone $z - 1$ sits behind every cut z does *plus* the cut separating $z - 1$ from z , so its set of relievable cuts contains that of z and its arbitrage value is no smaller: $\lambda_{z-1} \geq \lambda_z$, i.e. λ_z is non-decreasing up the cascade.

(iii) The budget $\sum_z k_z \leq K$ has the single multiplier $\mu \geq 0$; stationarity in $k_z \geq 0$ gives $\lambda_z \leq \mu$ with equality whenever $k_z > 0$ (complementary slackness). Raising K therefore allocates first to the zone of largest λ_z – the most upstream binding zone by (ii) – until its marginal value falls to the next, an interior optimum equalising $\lambda_z = \mu$ across the zones in use. The single-cut, single-period case has $\lambda = \pi_{\beta}$, the flexibility-equivalence of §7. $\square$

Appendix B. Sensitivity analysis

This appendix gives the full grids summarised in Figure 4 and the method behind them. Each sweep varies one lever of the faithful B6 two-zone instance (§5) over a grid, holding the others at their central calibration ($\bar{B} = 6.1$ GW,

central Scottish-wind and demand shares, no behind-boundary flexibility, $\mu = 0.30$), re-solves all 48 settlement periods of 8 December 2024, and aggregates to a daily total. The congestion rent R_{cong} is read from the boundary LP dual on the cost-reflective solve and the strategic component as the incremental cost $M = RC(\mu) - RC(0)$, so that $\Gamma = R_{\text{cong}} + M$ holds exactly. The driver is the routine `sensitivity.py`.

Table B.5: Full sensitivity grids (daily totals, £m), 8 December 2024. Central case in **bold**.

<i>(a) B6 transmission capability</i>						
$\bar{B}$ (GW)	4.0	5.0	6.1	7.0	8.0	9.0
R_{cong}	11.54	7.42	2.88	0.33	0.00	0.00
<i>(b) Scottish wind penetration (multiple of central output)</i>						
scale	0.8	1.0	1.2	1.4	1.6	
R_{cong}	0.00	2.88	8.88	14.88	20.88	
<i>(c) Flexibility relocated behind the boundary</i>						
ϕ (GW)	0.0	1.0	2.0	3.0	4.0	
R_{cong}	2.88	0.19	0.00	0.00	0.00	
<i>(d) Strategic markup μ</i>						
μ	0.00	0.15	0.30	0.45		
M	0.00	0.18	0.37	0.55		
Γ	2.88	3.07	3.25	3.43		

Panels (a) and (c) bracket the policy of transmission build-out and of siting flexibility behind the cut: both drive the relieved overload, and hence R_{cong} , to zero (at ≈ 8 GW of capability or ≈ 2 GW of flexibility), the network-feasible point of Proposition 2(i). Panel (b) shows the cost rising steeply and near-linearly with wind penetration above the binding threshold, quantifying the growth in the cost of national pricing along the Clean Power 2030 trajectory. Panel (d) isolates the strategic markup, linear in μ and additive on the congestion rent. The controlled welfare-loss experiment of §6.1, on the purpose-built convex/nonconvex instances with known costs (the role RTS-96 plays in Bichler et al. [5]), reproduces Propositions 1–2 and is driven by `nonconvex_experiment.py`.

Appendix C. Robustness of the Scotland/B6 classification

Because the realised annual figure rests on identifying the Scottish (north-of-B6) generation, and because transmission-connected units carry no GSP group in the BMRS registry, we classify a unit as Scottish if its GSP group is `_N/_P` or its zonal transmission loss factor (TLF) is at most -0.0055 . We verify that identification three ways; the driver is `robustness_classification.py`.

Threshold-invariance. GB TLFs are assigned by geographic zone and, for transmission-connected wind, take a small set of discrete values. Scotland occupies the two most-negative zones, -0.0200 (North Scotland) and -0.0066 (South

Scotland); the nearest value to the south is -0.0045 (the Cumbria/NW England zone – Walney, Robin Rigg, Morecambe). No transmission wind unit lies in the interval $(-0.0066, -0.0045)$, so every cut in that gap – including our -0.0055 – yields an identical Scottish set. Table C.6 confirms this numerically: cuts at -0.0045 , -0.0055 and -0.0065 return the same annual rent to the pound.

Sensitivity to the definition. Table C.6 also brackets the figure under deliberately distorted cuts. Admitting the Cumbrian zone (-0.0045) leaves it unchanged, because those units sit south of B6 and carry no B6 curtailment; restricting to North Scotland only (-0.0200), which wrongly drops genuine South-Scotland farms (Whitelee, Clyde, Crystal Rig), lowers it only to £1,008m. The headline is thus stable in [£1.0, £1.1]bn across any defensible definition. Only the GSP-group-only rule – which silently drops every transmission-connected farm, whose GSP field is blank – collapses it to £100m; that rule is the bug this classification corrects, not a competing definition.

Named cross-check. The curtailment is carried by the expected units. Across the sampled days, 123 distinct Scottish wind units are curtailed; the fifteen largest – led by Seagreen (T_SGRW0), Moray East (T_MOWE0), Dorenell and Viking – account for 69% of the volume and are, by their registry lead-party names, unambiguously transmission-connected Scottish wind farms. The classifier therefore selects the set an analyst would select by hand.

Table C.6: Annual realised B6 rent under alternative definitions of the Scottish (north-of-B6) set, on the same 30-day stratified estimator. The three TLF cuts spanning the Scotland–England gap are identical to the pound; the figure is stable across defensible definitions and collapses only under the incorrect GSP-only rule.

Scottish set definition	Rent (£m/yr)	Curtail. (TWh)	Share
GSP group only (pre-fix; incorrect)	100	0.67	5%
TLF ≤ -0.0200 (North Scotland only)	1,008	8.70	53%
TLF ≤ -0.0065 (in-gap)	1,126	9.55	59%
TLF ≤ -0.0055 (baseline)	1,126	9.55	59%
TLF ≤ -0.0045 (incl. Cumbria)	1,126	9.55	59%

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